

New Zealand vs Small Advanced Economy Research Priorities: A Corrected Reanalysis of the PMSITAC Report Findings

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Recommendations

- **Benchmarking of New Zealand's budget allocations for public R&D spending should be based on a complete picture of allocations in peer countries.** This should take into account public R&D funds that are not targeted or that are allocated at arms length from central government budgets.
- **Revisit the empirical case for the Technology for Prosperity funding increase.** The large changes in funding are currently justified partly on an 'underinvestment vs SAE peers', framing that doesn't hold up under corrected benchmarking. The investment may still be justified on other grounds (translating research into growth, strategic positioning, absorbing frontier technology), but those should be made explicitly rather than leaning on a comparator claim that doesn't survive scrutiny.
- **Give social sciences and humanities explicit consideration**, rather than leaving them uncovered by any of the four pillars. Underinvestment in social science and humanities relative to SAE peers is a robust finding and is evident in the original report's own raw numbers.
- **Decouple the agriculture and environment sciences in any future analysis.** The original report bundled these together under one reallocation rationale. Agriculture overweighting is robust; evidence for environmental sciences overweighting is much weaker than claimed. Treating them as a single "primary sector + environment" pot for reallocation risks cutting environmental science to an extent that the data doesn't support.
- **Invest in NZ's own current SEO/FORD reporting** rather than relying on decade-old GBARD figures or ad hoc MBIE internal classifications that don't map cleanly onto international standards. This would remove the need for crosswalk-based estimation entirely for future comparisons.
- **Any future reallocation decision at this scale should come with a sensitivity analysis**, not a point estimate. Given how much the headline comparison moved between methodologies here, a > \$100m decision resting on a single bar chart understates the real uncertainty.
- **Note the small-N caveat explicitly in any public communication.** Even the corrected 14-country SAE average is a small sample; a couple of unusual observations can move the benchmark meaningfully. Policy framed around "we lag SAE peers by X%" should carry visible confidence intervals, not bare point estimates.

1. Introduction

This note considers the data and analysis used in the recent report to the Prime Minister on “Prioritisation in New Zealand’s Science, Innovation and Technology system” or PMSITAC (Ministry of Business, Innovation and Employment, 2026). In particular, the report uses R&D expenditure data (sourced from OECD (2025) and Eurostat (2025)) to argue that New Zealand spends proportionately more on agricultural and environmental science than peer nations and underfunds high technology investments. Unfortunately the data used does not provide a full account of public R&D efforts in peer countries and because of this it misrepresents funding priorities in these nations. The purpose of this note is to examine New Zealand’s budget allocations for public R&D based on a more complete picture of public R&D allocations to account for institutional differences in funding allocation.

2. Comment on PMSITAC methodology

2.1 Use of GovERD

I begin by looking at the use of GovERD as a percentage of GDP in the PMSITAC report (Ministry of Business, Innovation and Employment, 2026, Figure 1). GovERD is the expenditure on R&D within the government sector (i.e. government labs and agencies, such as CRIs/PROs), as opposed to R&D expenditure in universities or businesses that might be funded by government. It is important to realise that the size of the government R&D sector varies considerably between countries with many relying more heavily on universities (via HERD) than government labs and agencies (GovERD). By proportion of GDP, New Zealand’s government sector is typical for the OECD (~0.25%), but as a proportion of total public expenditure on R&D, New Zealand is an outlier: GovERD represents 43% of public sector expenditure in New Zealand compared an average of 29% in the countries that PMSITAC used as a benchmark. All else being equal, New Zealand’s choices regarding its distribution of GovERD will tend to look unusual when measured as a proportion of GDP as the PMSITAC report does, because we invest a relatively large proportion of public funding in GovERD compared to our peers.

Indeed, one of the reasons New Zealand appears to devote considerably more GovERD to the environment than other countries is because we happen to have a significant government lab focussed on environmental science (e.g. formerly Manaaki Whenua - Landcare Research, now part of the Bioeconomy Research Institute). This makes comparison with other countries difficult. Countries with smaller government research sectors may fund more environmental science than New Zealand via their university sectors, for instance, but this would not be apparent in GovERD data. In other words, differences in GovERD may say as much about institutional arrangements for R&D as they do about priorities. At best this evidence tells us that New Zealand conducts a relatively large amount of agricultural and environmental science in its government labs compared to OECD peers. We conclude that GovERD as a percentage of GDP can’t be used on its own to make inferences about national priorities in Small Advanced Economies (SAEs).

2.2 Use of GBARD

A second issue with the report lies in how it uses GBARD: government budget allocations for R&D (Ministry of Business, Innovation and Employment, 2026, Figure 2). GBARD covers all government budget allocations for R&D so, in principle, it is a better metric for comparing spending priorities than GovERD. However, the GBARD data reported in the report does not

provide a full representation of overall national priorities. For instance, the report suggests that Switzerland - a country known particularly for its health research - devotes only ~1% of GBARD to health R&D, but this simply reflects the fact that Switzerland prioritises a relatively small amount of its public R&D funding at the central government level. In fact only 12% of Swiss GBARD is allocated by its national government to particular sectors at budget¹ and it is only this portion that is disaggregated in the PMSITAC report. The other 88% of Swiss GBARD is characterised as “general advancement of knowledge” (GAK) and either goes to university research or is allocated by independent agencies such as its National Science Foundation. By omitting this, the PMSITAC report gives the impression that health research is not a priority in Switzerland, when the reality is that Switzerland’s key priorities are largely set at arm’s length from the government budget process. Indeed, Switzerland reports 23.1% share for health and medical research in its public Field of Research and Development data (OECD 2023). I conclude that the GBARD SEO data used is a poor representation of national priorities and should not be used as a benchmark for New Zealand’s research system.

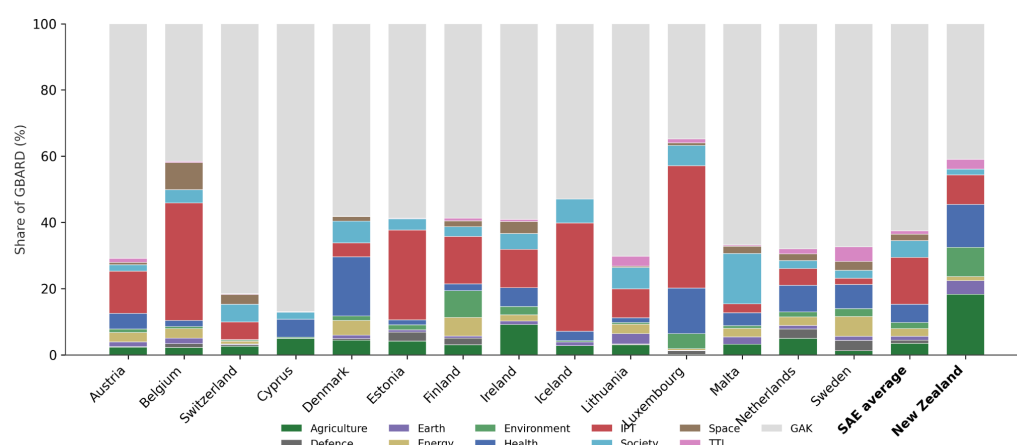


Figure 1: GBARD data including GAK for a selection of SAEs . For New Zealand I use GAK data from 2011-17; New Zealand ceased reporting GBARD data by SEO in 2017, so GAK is only available up to this date. The figure shows that New Zealand prioritises a relatively large proportion of its public R&D funding at central government level (Source: OECD, 2023).

3. Improved Methodology

To understand how the incomplete use of GBARD data affects PMSITAC’s proposed priorities we need to compare New Zealand’s GAK to the benchmark SAEs (Ministry of Business, Innovation and Employment, 2026). For most countries GAK is exactly the sum of non-oriented research funding (NOR) and general university funding (GUF), which I denote as GAK* (this doesn’t seem to be the case for France and Luxembourg). As is shown below in Figure 2, New Zealand has a share of GAK* well below the SAE average. It is likely, therefore, that the approach used in the PMSITAC report, which neglects GAK, will introduce bias into the comparisons made. It is also interesting that New Zealand’s central government prioritises a much larger share of public R&D funding that most other SAEs.

¹ Eurostat 2025

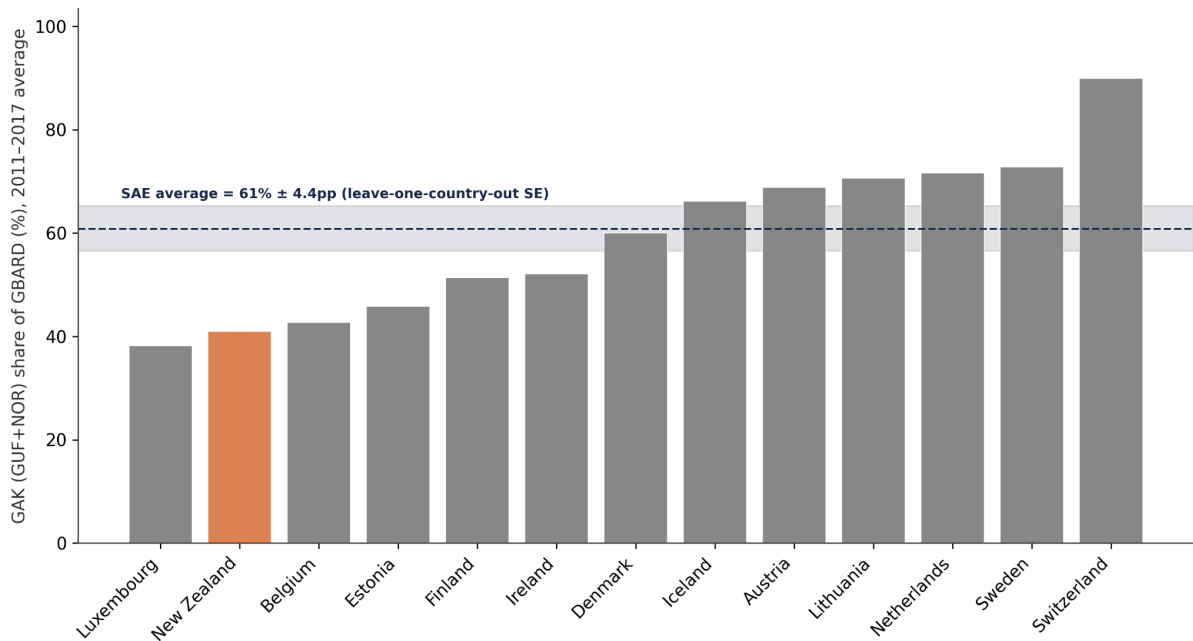


Figure 2: $GAK^* = (GUF+NOR)/GBARD$ estimated for the set of small advanced economies used as comparators in the PMSITAC report. For New Zealand I use GAK^* from 2011-17; New Zealand ceased reporting NOR and GAK in 2017. The standard error about the SAE average (dotted line) is computed using jackknife resampling. The figure shows that New Zealand prioritises a relatively large proportion of its public R&D funding at central government level (Source: OECD, 2023).

Correcting for this is not straightforward. The approach I adopt here is to analyse New Zealand's GBARD with GAK^* (both estimated from 2011-17 data) to give a fuller picture of its national funding landscape. I then map the results into FORD shares, which can be compared with other SAEs. The mapping from GBARD to FORD uses a crosswalk matrix that distributes each GBARD category (including GAK^*) across FORD fields (natural sciences, engineering, medical, agricultural, social, humanities) using weights that reflect a derived mapping from a 32-country panel of countries that report both GBARD by SEO and FORD (see Appendix). This produces an estimate of the implied distribution of research effort by field in New Zealand, allowing consistent comparison across countries despite differences in institutional structure and classification. This approach should reduce the distortions caused by assuming GBARD represents complete national research priorities, providing a more faithful representation than the use of raw GBARD. Finally I compare New Zealand's derived FORD shares with those from SAEs that reported FORD in 2023 (AT, BE, CH, CY, DK, EE, FI, IE, IS, LT, LU, MT, NL, SE) with standard errors computed using jackknife resampling.

The result is shown in Figure 3. The key findings are:

- Agriculture is robustly overweighted: NZ spent ~3 times the SAE average, and this would fall to ~2.4 under the PMSITAC proposed reweighting.
- Social sciences and humanities are both dramatically underweighted in NZ (social sciences at 39% of the SAE share; humanities at 21%).
- Natural sciences finds NZ just ~15% above the SAE average, which makes a large overweighting of environmental science less likely.
- NZ's Engineering & Technology share is indistinguishable from the SAE with this method and so not meaningfully underweighted. But the *PMSITAC-proposed*

reallocation would push NZ to 0.28, about 28% above the SAE average, a larger correction than closing any real gap would require.

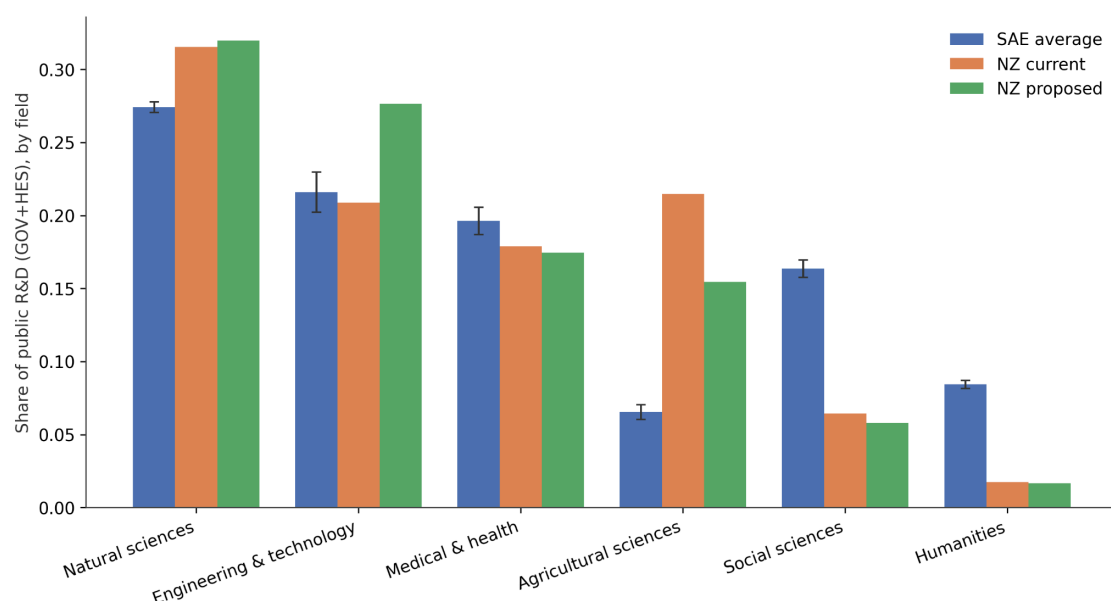


Figure 3: A corrected comparison of research priorities by FORD in peer SAEs, and in New Zealand (both actual and proposed) as mapped from New Zealand's 2011-17 GBARD SEO shares using the crosswalk matrix (see Appendix). The standard errors for the SAE averages are computed using jackknife resampling ('leave-one-country-out'). (Source: OECD 2025, Eurostat 2025).

Discussion

Here we have benchmarked New Zealand's budget allocations for public R&D spending using a methodology that gives a complete picture of public R&D in peer countries. This method takes into account public R&D funds that are not targeted or that are allocated at arms length from central government budgets, but does so at the expense of losing resolution (e.g it can't separate environmental science from natural sciences). It also assumes that New Zealand's GAK is typical, which it may not be given the idiosyncrasies of its historic funding distribution². Furthermore, there is a potentially important distinction to make between FORD (the 'what') and SEO (the 'why'): that is, the analysis here compares the type of research taking place in peer countries, not necessarily why they are doing it.

With these caveats, some of the findings of the PMSITAC report are found to be robust, but others are not. In particular, the PMSITAC finding that New Zealand overweights agricultural sciences relative to its peers is robust. However, the claim that environmental sciences is overweighted in New Zealand is much weaker. Treating them as a single "primary sector + environment" pot to redirect funding away from risks cutting environmental science with no evidential basis. The findings here also substantially weaken the empirical case for the funding increase for Technology for Prosperity: New Zealand does not seem to underinvest in these areas relative to its SAE peers. The investment may still be justified on other grounds (e.g. translating research into growth, strategic positioning, absorbing frontier technology), but these arguments and the trade-offs involved (weaker stewardship of biodiversity, for instance) should be made explicitly rather than leaning on a comparator claim that doesn't survive scrutiny.

² This could be partly mitigated through the use of bibliometrics, for instance.

Perhaps the strongest finding of this reanalysis is the underweighting of social sciences and humanities in New Zealand. This is a robust finding with the method used here and is evident in the original report's own raw numbers. Given the strong discrepancy with SAE peers this should be given explicit consideration, rather than leaving them uncovered by any of the four pillars. This is especially important given the removal of these areas from the Marsden Fund and the historical under-funding (Gordon, 2024).

One of the findings is that New Zealand's public R&D system is more centralised than the typical SAE (at least when using New Zealand's 2011-17 data). This centralisation may well have played a role in preserving the unusual funding distribution revealed in the analysis here. Furthermore, it is arguable that many of the recommendations in the PMSITAC report (e.g. the four pillars) will increase that centralisation making New Zealand's public R&D system even more of an outlier and perpetuating what has been a highly idiosyncratic funding distribution.

Finally any future reallocation decisions at this scale should come with a sensitivity analysis, not a point estimate: a > \$100m decision resting on a single bar chart understates the real uncertainty. Policy framed around "we lag SAE peers by X%" should carry visible confidence intervals, not bare point estimates. Finally New Zealand should invest in current SEO/FORD reporting rather than relying on decade-old GBARD figures or ad hoc MBIE internal classifications that don't map cleanly onto international standards. This would remove the need for crosswalk-based estimation entirely for future comparisons.

Summary

In summary, the quantitative benchmarking presented in the PMSITAC report is not sufficiently robust to support the scale and direction of the proposed reallocation. Once institutional differences in research funding are accounted for, some PMSITAC conclusions survive and while others change substantially. Agriculture remains overweight; the apparent technology deficit largely disappears; a major deficit in funding for social science and humanities becomes even more apparent; and New Zealand's funding system is revealed to be substantially more centralised than its SAE peers. We conclude that a comparative analysis with the SAEs does not support the proposed > \$100m reallocation in research funding.

References

Eurostat. (2025). *Research and development statistics*. European Commission.
<https://ec.europa.eu/eurostat/web/science-technology-innovation/data/database>

Ministry of Business, Innovation and Employment. (2026). *Report to the Prime Minister: Prioritisation in New Zealand's science, innovation and technology system*. New Zealand Government. [MBIE report](#).

Organisation for Economic Co-operation and Development. (2025). *Main Science and Technology Indicators (MSTI) database*. OECD. <https://doi.org/10.1787/data-00182-en>

Gordon, P., and Argue, M. (2024, December 4). *Government's Marsden Fund cuts: All humanities, social sciences research funding slashed*. RNZ.

Appendix

The SEO to FORD crosswalk matrix was estimated empirically using Eurostat data for the 32 countries reporting both GBARD-by-socio-economic-objective and GERD-by-field-of-research (restricted to GOV+HES performing sectors) in 2023. We solved for the SEO×FORD matrix W that minimises the squared error between each country's actual reported FORD shares and its SEO shares projected through W , subject to each row of W being non-negative and summing to one. To prevent overfitting given the panel size relative to the 66 free parameters (11 SEO categories × 6 FORD categories), the objective included an L2 penalty shrinking W toward an initial guess, which served as both a regularisation prior and a benchmark. The fitted matrix was validated by leave-one-country-out cross-validation, which showed it out-predicted the static heuristic matrix out-of-sample (mean absolute error 0.047 vs. 0.055), before being applied to translate New Zealand's own SEO-classified budget shares into implied field-of-research shares for comparison against the small-advanced-economy benchmark (Figure 3, main text). Note that the wider 32-country panel is used only to fit the crosswalk; the SAE benchmark itself is the average of the 14 countries that meet the SAE definition (following PMSITAC this is population <20m, per-capita income >US\$30k), namely AT, BE, CH, CY, DK, EE, FI, IE, IS, LT, LU, MT, NL, SE and report suitable data. Note that Australia and Israel were excluded as they don't consistently report SEO and/or FORD data.

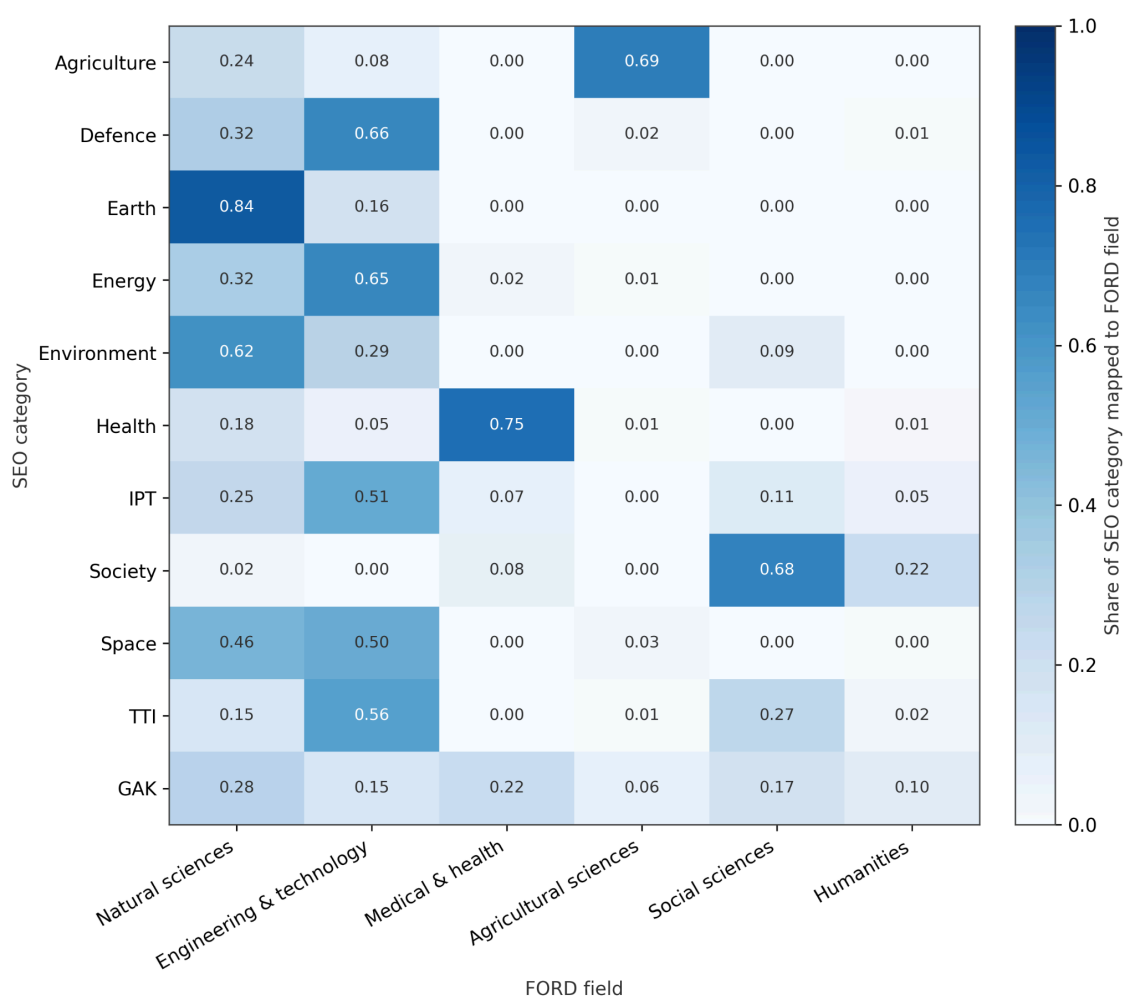


Figure A1: Cross-walk matrix derived from a 32-country panel comprised of those countries that report both FORD and GBARD by SEO (Source: Eurostat 2025).